

Olympiad Test Paper — Social Science (Class 7) — Paper 4

Chapter 13: The Story of Indian Farming

Paper 4 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	13 — The Story of Indian Farming
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: most rural farm work is done by women not men; the Green Revolution had a MIXED record (more grain but degraded soil, depleted groundwater, pollution, health harm, debt); rabi is the dry winter crop that usually needs irrigation while kharif lives on the monsoon; intensive/extensive is about effort per area, not purpose; subsistence/commercial is about purpose, not plot size; organic farming cuts chemicals rather than abandoning food; hydroponics shows plants can grow without soil. Do not write on this paper — mark answers on your answer sheet.

1. A farmer keeps almost the whole harvest to feed her household, works her single hectare very intensively with much hand labour and water, and sells nothing. Which TWO labels fit her at once?

- (A) commercial (by purpose) and intensive (by effort per unit area)
- (B) subsistence (by purpose) and intensive (by effort per unit area)
- (C) subsistence (by purpose) and extensive (by effort per unit area)
- (D) commercial (by purpose) and extensive (by effort per unit area)

2. Which pairing of a farming term with its defining axis is the ODD ONE OUT (mismatched)?

- (A) subsistence farming -> classified by the purpose of the crop (eat or sell)
- (B) commercial farming -> classified by the purpose of the crop (eat or sell)
- (C) intensive farming -> classified by the purpose of the crop (eat or sell)
- (D) extensive farming -> classified by the effort applied per unit of land

- 3.** A grower raises only fruits, vegetables and flowers near a city and trucks the entire output to hotels for cash. Two labels apply together. They are:
- (A) horticulture and subsistence farming
 - (B) horticulture and commercial farming
 - (C) cereal farming and commercial farming
 - (D) terrace farming and subsistence farming
- 4.** Over three years the SAME field is called subsistence, then commercial, then subsistence again by different observers. The single fact that best explains this without anything physical changing is that:
- (A) the soil type of the field changed each year
 - (B) the field's cropping season switched between kharif and rabi
 - (C) the label tracks the farmer's purpose, which can shift between feeding the family and selling for income
 - (D) intensive and extensive farming mean the same as subsistence and commercial
- 5.** A landless woman is hired by the day to transplant and weed on a large farm that sells all its grain. She owns no land and keeps no crop. Within the chapter's picture she is best understood as:
- (A) a subsistence farmer feeding her own family
 - (B) a commercial farmer selling her own grain
 - (C) outside agriculture entirely because she owns no land
 - (D) essential farm labour, the input on which Indian agriculture heavily depends
- 6.** A development officer claims, 'Commercial farming is always better for the nation than subsistence farming.' The most chapter-accurate reply is that:
- (A) each suits different families and conditions; commercial farming earns income but raises cost and market risk, while subsistence farming secures a family's food
 - (B) yes, because selling crops is always more valuable than eating them
 - (C) no, because subsistence farming always produces more food than commercial farming
 - (D) the two terms are identical, so the comparison is meaningless
- 7.** Farmer P works a large holding thinly, sowing one hardy crop and waiting on the rain for a low yield per hectare. Which single pair of labels best fits Farmer P?
- (A) intensive and irrigated
 - (B) extensive and irrigated
 - (C) intensive and rain-fed
 - (D) extensive and largely rain-fed
- 8.** Which statement BEST explains why 'all Indian farming is basically the same' is a mistaken claim?
- (A) farming varies at once by soil, climate, season, water source and purpose, so practices differ widely across regions
 - (B) because every Indian farm uses exactly the same seeds and the same tools
 - (C) because every Indian farm is commercial and fully irrigated
 - (D) because Indian farming has never changed since ancient times

- 9.** Place these three events in correct order through one year starting just after spring: the cool winter crop, the short hot in-between crop, the monsoon crop. The right sequence is:
- (A) cool winter (rabi), then short hot in-between (zaid), then monsoon (kharif)
 - (B) monsoon (kharif), then cool winter (rabi), then short hot in-between (zaid)
 - (C) monsoon (kharif), then short hot in-between (zaid), then cool winter (rabi)
 - (D) short hot in-between (zaid), then monsoon (kharif), then cool winter (rabi)
- 10.** A crop is sown with the first heavy June rains and harvested as those rains end and the cool season begins. The season and the most fitting water source are:
- (A) rabi crop, usually needing winter irrigation
 - (B) zaid crop, grown in the short hot gap between seasons
 - (C) rabi crop, able to live mainly on monsoon rain
 - (D) kharif crop, able to live mainly on monsoon rain
- 11.** Why do many RABI crops in a dry-winter belt need irrigation while many KHARIF crops in the SAME belt usually do not?
- (A) kharif crops fall in the wet monsoon while rabi crops grow in the comparatively dry winter, so rabi must be watered artificially
 - (B) rabi crops dislike water, so irrigation is only used to trick them into growing
 - (C) the monsoon rains in that belt fall mainly during the winter
 - (D) kharif crops there are grown only with canal water, not rain
- 12.** A farmer in a region served mainly by the NORTHEAST monsoon times her main crop to its arrival. The heaviest rain she can count on falls around:
- (A) June to September
 - (B) March to May
 - (C) October to December
 - (D) evenly through all twelve months
- 13.** Two adjacent farms grow the same kharif crop in the same year. Farm A is rain-fed only; Farm B has tube wells and canals. A late, weak monsoon arrives. The MOST likely outcome is that:
- (A) Farm B is hit hardest because pumps need rain to operate
 - (B) Farm A is hit hardest because the monsoon is its only water source, while Farm B can fall back on its wells and canals
 - (C) both farms are affected equally because they grow the same crop
 - (D) Farm A benefits because rain-fed crops grow better in a dry year
- 14.** A farmer insists on sowing wheat (a rabi crop) in June as the southwest monsoon arrives. The MOST likely result, by the chapter's logic, is that:
- (A) it thrives, because every crop grows equally well in every season
 - (B) the crop struggles, because wheat suits the cool, drier winter rather than the hot, wet monsoon
 - (C) it becomes a kharif crop and yields more than ever before
 - (D) it turns into a zaid crop and ripens within two weeks

- 15.** A government wants to cut crop failures in years when the monsoon is unreliable. The MOST directly relevant investment from the chapter is to:
- (A) expand irrigation such as canals, wells and drip systems so crops get water even in weak-monsoon years
 - (B) ban all kharif crops so farming happens only in the winter
 - (C) push every farm to depend more heavily on rainfall alone
 - (D) require every field to grow a single crop every year
- 16.** A farmer wants a quick crop in the brief hot window AFTER he harvests his winter crop but BEFORE he sows his monsoon crop. He is planning a:
- (A) rabi crop, grown through the long cool winter
 - (B) kharif crop, sown with the arriving southwest monsoon
 - (C) zaid crop, grown in the short summer between rabi and kharif
 - (D) perennial crop that grows for several years at a stretch
- 17.** Which single statement about the three cropping seasons is FALSE?
- (A) kharif crops are sown with the arriving southwest monsoon in the hot, rainy season
 - (B) rabi crops grow in the cool winter with less rain and are harvested in spring
 - (C) zaid crops are short crops grown in the hot gap between rabi and kharif
 - (D) rabi is the monsoon season's main crop, sown in June and living on heavy rain
- 18.** Two neighbouring fields grow the same crop, but one sits on fertile river silt and yields far more. The BEST single explanation is that:
- (A) the richer field's soil is actually wind-blown desert sand
 - (B) fertile soil needs no nutrients at all to feed the plants
 - (C) river silt blocks the roots and therefore lowers the yield
 - (D) alluvial soil, made of silt brought by rivers, is rich in nutrients and holds moisture well
- 19.** A farmer mixes plenty of well-rotted compost and manure into a tired field, and the next crop improves. The BEST explanation is that the humus has:
- (A) added nutrients and helped the soil hold moisture, making it more fertile
 - (B) removed all water from the soil so the roots stay dry
 - (C) hardened the loose soil into solid rock
 - (D) killed the soil organisms that the crop depends on
- 20.** A field grows the same nutrient-hungry crop every year and the yield slowly falls. A pulse crop is rotated in, and the following cereal grows better. The reason rotation helped is that:
- (A) different crops draw on and return different nutrients, so the soil keeps a better nutrient balance
 - (B) rotation physically spins the field to mix the soil layers

- (C) rotation removes every nutrient from the soil at the same time
- (D) rotation stops crops from ever growing in that field again

21. On a steep hillside a farmer cuts the slope into level steps before planting. The TWO benefits that follow are best described as:

- (A) the steps remove the need for any soil or any water
- (B) the steps turn the hillside into barren desert sand
- (C) the steps hold rainwater and reduce soil being washed downhill, letting crops grow on steep land
- (D) the steps let crops grow without using any seeds

22. A farmer relies only on the monsoon and has no canal, well, pump or drip line. By the chapter's logic, the SINGLE biggest risk to her harvest is:

- (A) a weak or late monsoon, since rain is her only water source
- (B) too much canal water flooding her field
- (C) her irrigation pump breaking down
- (D) her drip-system emitters becoming clogged

23. Why is a low raised embankment (a bund) built around or across a field useful to the farmer?

- (A) it holds water in the field and helps stop soil from being washed away
- (B) it is a high-yield seed that makes crops grow faster
- (C) it is a chemical that kills the field's pests
- (D) it is a short summer crop grown between seasons

24. India's many old tanks, bunds and step-wells, standing alongside modern canals and tube wells, BEST show that managing water for crops is:

- (A) a problem first invented only after the Green Revolution
- (B) a long-standing concern that farmers have met with both traditional and modern methods
- (C) something Indian farmers historically never thought about
- (D) needed only in regions that receive no rain at all

25. A farmer weighs saving seeds from her own harvest against buying high-yield seeds from a company. The MOST balanced, chapter-consistent reasoning is that:

- (A) saved seeds cost little and suit local conditions, while bought high-yield seeds can lift yields but cost money and often need more water and chemicals
- (B) saved seeds always out-yield every modern seed in every field
- (C) bought seeds are free and never need extra water or chemicals
- (D) the choice makes no difference because all seeds behave identically

26. Why might a farmer deliberately COMBINE saved local seeds and compost with a careful, modest use of some modern inputs?

- (A) to make farming as harmful to the land as possible
- (B) to get reasonable yields while protecting the soil and keeping costs down
- (C) because a law forbids the use of any modern tool
- (D) to prove that chemicals alone can never raise yields

27. Which of the following is NOT one of the inputs the Green Revolution package combined to raise food output?

- (A) high-yielding seed varieties
- (B) a switch to barter trade in place of money
- (C) irrigation
- (D) chemical fertilisers and pesticides

28. A claim says, 'Traditional saved-seed methods always produce more food than the Green Revolution package ever could.' The MOST accurate correction is that:

- (A) the claim is right, because saved seeds always out-produce high-yield seeds
- (B) the Green Revolution sharply raised food-grain output and helped India feed itself, even though it brought serious long-term harms
- (C) the Green Revolution actually lowered India's food output
- (D) neither approach has any effect on the amount of food produced

29. Heavy, year-after-year pumping for irrigation in a Green-Revolution belt is MOST likely, over time, to lead to:

- (A) rising groundwater that floods the whole region
- (B) no change in groundwater, because pumping does not affect it
- (C) falling groundwater levels that threaten future water supply for crops and people
- (D) the creation of brand-new underground rivers

30. Assertion: The Green Revolution raised India's food-grain output. Reason: It did so because it had no harmful side effects of any kind. Which is correct?

- (A) both the assertion and the reason are true, and the reason explains the assertion
- (B) both the assertion and the reason are false
- (C) the assertion is true but the reason is false; output rose, yet the Green Revolution also caused real harms
- (D) the assertion is false but the reason is true

31. A region that LED the Green Revolution now reports tired soil, low water tables and farmers struggling to repay loans for seeds and chemicals. The BEST single explanation is that:

- (A) these troubles are unrelated to how the land was farmed
- (B) the same intensive, input-heavy methods that raised yields also degraded soil and water and pushed up costs, so the gains came with long-term costs
- (C) the region simply never adopted the Green Revolution at all
- (D) the debt was caused by low yields rather than by high input costs

32. Years of heavy chemical fertiliser and pesticide use on an intensively farmed field are MOST likely, as a cause-and-effect chain, to:

- (A) pollute soil and water and degrade the soil's natural fertility over time
- (B) make the soil grow steadily richer and cleaner each year
- (C) let the chemicals vanish without ever touching soil or water
- (D) increase the local rainfall because of the chemicals

33. A student writes that the Green Revolution was 'an unmixed success.' The single BEST correction, true to the chapter, is that it:

- (A) actually reduced India's overall food supply
- (B) raised food output but also impoverished soil, depleted groundwater, polluted soil and water, harmed health and left many farmers in debt
- (C) had no measurable effect on farming of any kind
- (D) benefited the soil and groundwater while lowering yields

34. An organic farm and a chemical-intensive farm both grow wheat side by side. The CLEAREST single difference between them is that the organic farm:

- (A) grows its wheat entirely without any soil
- (B) produces no wheat at all
- (C) uses far more chemical pesticide than the other farm
- (D) avoids synthetic fertilisers and pesticides in favour of natural inputs like compost, manure and neem

35. Why might shifting a chemical-heavy farm toward organic and natural methods improve the local groundwater over many years?

- (A) because organic farming uses no water and needs no soil at all
- (B) because natural inputs magically refill the aquifer overnight
- (C) because organic crops do not grow, so no water is ever used
- (D) cutting chemical inputs and easing back on heavy irrigation lessens pollution and slows groundwater depletion

36. Why does the chapter present organic and natural farming as a 'RESPONSE' rather than as the original way farming was done?

- (A) because farming before the Green Revolution always relied on heavy chemicals
- (B) they are deliberate moves to cut chemical harm after the Green Revolution exposed that harm
- (C) because organic methods were invented in the 1960s alongside the chemicals
- (D) because natural inputs like manure and neem were unknown until very recently

37. Which definition BEST captures 'sustainable farming' as the chapter uses the term?

- (A) farming that meets today's needs without harming the soil, water or the ability to keep farming in future
- (B) farming that uses up every resource as fast as possible

- (C) farming that produces no food at all in the present
- (D) farming that ignores the future entirely to maximise this year's profit

38. Keeping soil productive through crop rotation, adding humus and preventing erosion is COLLECTIVELY called:

- (A) soil irrigation
- (B) soil conservation
- (C) soil mechanisation
- (D) soil export

39. A farmer worries that switching FULLY to organic methods all at once may lower her income in the first years. The most reasonable, chapter-consistent path is to:

- (A) blend natural methods with a careful use of some modern inputs, balancing yield, soil health and cost during the transition
- (B) use the heaviest possible dose of chemicals to be safe
- (C) stop farming entirely until organic prices rise
- (D) ignore soil health and chase the maximum yield every year

40. A teacher claims, 'A pesticide is a kind of water sprayed to help crops drink.' Using the meaning of the '-cide' ending, the correct statement is that a pesticide:

- (A) waters the crops, because '-cide' means 'water'
- (B) is a seed, because '-cide' means 'seed'
- (C) kills crop pests, because '-cide' means 'killer'
- (D) is a garden, because '-cide' means 'garden'

41. A claim says, 'Plants can only ever grow in soil.' The chapter shows this is wrong because:

- (A) plants actually need no nutrients of any kind to grow
- (B) plants can only grow on bare rock, never in soil
- (C) plants can grow only in the desert and nowhere else
- (D) hydroponics lets plants grow in nutrient-rich water, with minerals supplied directly to the roots

42. Why is hydroponics described as especially useful in particular settings?

- (A) it lets crops grow where there is no water at all
- (B) it removes the plant's need for any nutrients
- (C) it suits urban farming and places with poor soil, since it grows plants in nutrient water rather than soil
- (D) it works only for crops grown in ancient times

43. A survey records WHO actually sows, transplants, weeds and harvests on small farms across rural India. It would most likely find that the majority of this hands-on work is done by:

- (A) men only
- (B) machines only
- (C) women

(D) hired workers brought in from other countries

44. An economist notes that much of the field work done by women in rural India is UNPAID family labour that never appears in wage records. The most accurate conclusion is that:

- (A) women therefore do very little real farm work
- (B) only paid male labour should count as genuine farming
- (C) women's contribution to farming is often undercounted, even though they do a large share of the actual work
- (D) farming in modern India needs almost no labour at all

45. A new programme offers training and credit to 'farmers' but quietly assumes every farmer is a man who owns land. Why might this poorly serve rural farming?

- (A) because women contribute nothing to farming and need no support
- (B) because all farmers are in fact wealthy landowners anyway
- (C) because farm labour is no longer needed anywhere in India
- (D) it overlooks the many women and landless labourers who do much of the actual farm work

46. A claim says, 'Since tractors and machines now do most field work, women no longer play a role in Indian farming.' The BEST correction is that:

- (A) yes, machines now do everything, so no person works the fields
- (B) women never did much farm work to begin with
- (C) men have always done all the farm work in India
- (D) Indian farming is not fully mechanised, and women still do most of the agricultural work

47. Recognising that women carry out most farm operations should MOST reasonably lead agricultural programmes to:

- (A) stop offering any support to farms where women work
- (B) assume women will not benefit from better tools
- (C) direct training, tools and support to the women who actually do the work
- (D) replace all the women workers with imported labour

48. Mechanisation, such as using tractors and threshers, MAINLY helps farmers by:

- (A) removing the need for soil, seeds and water altogether
- (B) guaranteeing that every farmer becomes wealthy
- (C) making farming possible only during the winter
- (D) doing heavy field tasks faster and with less manual effort, especially on larger holdings

49. A small farmer with a tiny, fragmented plot is unsure whether to buy a large tractor. The MOST balanced view from the chapter is that mechanisation:

- (A) is equally easy and cheap for every farm regardless of size
- (B) is impossible to use anywhere in India
- (C) suits larger holdings better and may be costly or hard to use on very small, fragmented plots
- (D) removes the need for any human labour on every farm

50. A farmer brings in a good harvest yet earns very little because the price collapses when everyone sells at once. This BEST illustrates which challenge facing farmers?

- (A) the uncertainty of market prices, which can leave even a good harvest unprofitable
- (B) a weak monsoon that ruined the crop
- (C) soil so poor it produced no harvest at all
- (D) a complete absence of any buyers in the country

51. Erratic rainfall, sudden heatwaves and shifting season timings are making it harder for farmers to plan sowing and harvest. The challenge this MOST directly names is:

- (A) an excess of irrigation water everywhere
- (B) the soil suddenly becoming too fertile
- (C) climate change, which disrupts the timing and reliability of seasons and rainfall
- (D) the overnight disappearance of all farm labour

52. Across generations, family land is divided among heirs until each farmer holds only a small scattered patch. This trend CHIEFLY worsens farming by:

- (A) enlarging each farm so machines become essential
- (B) improving the soil fertility on every separate patch
- (C) guaranteeing a higher market price for the crop
- (D) shrinking and fragmenting holdings, making efficient farming and machinery use harder

53. Which set BEST names the real, present-day challenges facing Indian farmers as the chapter describes them?

- (A) abundant land, perfect prices and no climate of any kind
- (B) shrinking and fragmented landholdings, climate change, uncertain market prices, water stress and debt
- (C) a complete absence of any difficulty whatsoever
- (D) having no crops at all to grow anywhere in the country

54. A region adopted the full Green-Revolution package and now reports BOTH higher grain output AND degraded soil with falling water tables. The chapter's overall LESSON about boosting yields with chemicals is that it can:

- (A) bring only benefits and never any cost
- (B) bring only costs and never any benefit
- (C) bring short-term gains in output but long-term costs to soil, water and health
- (D) have no connection at all to the state of the soil or water

55. A farmer treats a pest outbreak with a neem-based spray instead of a synthetic chemical pesticide. This single choice is an example of:

- (A) the Green Revolution package in action
- (B) commercial export farming by its very definition
- (C) abandoning pest control altogether
- (D) organic farming, using a natural pest control in place of a chemical one

56. Which single statement, holding several threads together, BEST summarises the chapter's corrected picture of Indian farming?

- (A) the Green Revolution was purely good, and the typical farmer is a man using only chemicals
- (B) the Green Revolution had mixed effects, most rural farm work is done by women, and many farmers wisely blend traditional and modern methods
- (C) all Indian farming is identical and has never changed over time
- (D) women do no farm work and machines now do everything

57. Evidence of very early rice cultivation in India, dated to roughly the 7th-8th millennium BCE, and the continuation of farming right up to today TOGETHER best show that Indian agriculture is:

- (A) a modern invention only a few decades old
- (B) an ancient and continuous tradition
- (C) something India borrowed entirely from abroad
- (D) practised only in ancient times and then abandoned

58. A teacher says India has just ONE soil type, the same everywhere. Knowing India has several major soils such as alluvial, black and red, the BEST correction is that:

- (A) India has several major soil types, and what grows well varies from place to place with the soil
- (B) every field in India shares one identical soil
- (C) soil type makes no difference to which crops grow
- (D) only one single crop can be grown anywhere in India

59. Rice in the flooded paddies of the rainy season and wheat sown in the cool, drier winter are classic examples of, respectively:

- (A) a kharif crop and a rabi crop
- (B) a rabi crop and a kharif crop
- (C) a zaid crop and a kharif crop
- (D) a rabi crop and a zaid crop

60. Historians call the work women do the 'backbone' of rural farm operations CHIEFLY because they:

- (A) own most of the country's farmland
- (B) alone operate all the heavy farm machinery
- (C) manage farms only from offices and never enter the fields
- (D) perform the bulk of the hands-on tasks like sowing, transplanting, weeding and harvesting

Solutions — Social Science (Class 7), Chapter 13: The Story of Indian Farming — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	A	21	C	31	B	41	D	51	C
2	C	12	C	22	A	32	A	42	C	52	D
3	B	13	B	23	A	33	B	43	C	53	B
4	C	14	B	24	B	34	D	44	C	54	C
5	D	15	A	25	A	35	D	45	D	55	D
6	A	16	C	26	B	36	B	46	D	56	B
7	D	17	D	27	B	37	A	47	C	57	B
8	A	18	D	28	B	38	B	48	D	58	A
9	D	19	A	29	C	39	A	49	C	59	A
10	D	20	A	30	C	40	C	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Two independent axes apply. Purpose: she grows to feed her family and sells nothing, so it is subsistence, not commercial. Effort per area: she pours heavy labour and water onto one hectare, so it is intensive, not extensive. 'Commercial and intensive' wrongly flips the purpose to selling. 'Subsistence and extensive' wrongly calls heavy per-area effort thin/extensive. 'Commercial and extensive' is wrong on both counts.

2. (C) Intensive (and extensive) farming is defined by how heavily land is worked per unit area, not by purpose, so pairing 'intensive' with the eat-or-sell axis is the mismatch. Subsistence and commercial are correctly defined by purpose (eating vs selling). Extensive is correctly defined by effort per unit of land. Only the intensive pairing crosses the wrong axis.

3. (B) Cultivating fruits, vegetables and flowers is horticulture, and selling the whole output for cash makes the purpose commercial, so both apply. Subsistence is wrong because she keeps nothing to eat. 'Cereal farming' is wrong because she grows no grains.

'Terrace farming' describes step-cut hillsides, which is not stated, and subsistence is again wrong.

4. (C) Subsistence versus commercial is decided by purpose, and purpose can change year to year, so the same unchanged field can earn either label. Soil type does not flip annually. Season (kharif/rabi) is a different axis and does not set the subsistence/commercial label. Intensity (intensive/extensive) is a separate axis from purpose, so it is not a synonym for subsistence/commercial.

5. (D) She supplies the labour input that Indian farming relies on, and in rural India most such work is done by women. She is not a subsistence farmer because she keeps none of the crop, and not a commercial farmer because she owns and sells nothing. Far from being outside agriculture, landless labourers are central to the harvest, so 'irrelevant because landless' is false.

6. (A) The chapter treats the two as suited to different situations, not one as universally superior: commercial farming brings cash but also cost and market risk, while subsistence farming guarantees a family's own food. 'Selling always beats eating' ignores food security. Subsistence farms need not out-produce commercial ones. The two differ in purpose and are not identical, so the comparison is meaningful.

7. (D) Spreading limited effort thinly over a large area for a low per-hectare yield is extensive, and depending on the rains makes it rain-fed, so both apply. 'Intensive' is wrong because effort per area is low. 'Irrigated' is wrong because he waits on rain rather than supplying water artificially. Each alternative gets at least one axis wrong.

8. (A) Indian farming varies along several axes together, soil, climate, season, water source and purpose, so practices are far from uniform. Saying every farm uses identical seeds and tools, or that all farms are commercial and irrigated, merely restates uniformity in another form, which is false. And farming has changed greatly, including the Green Revolution, so 'never changed' is also wrong.

9. (D) After spring comes the short hot zaid window, then the southwest monsoon brings kharif, and after the rains the cool rabi winter follows: zaid -> kharif -> rabi. Starting with rabi or placing zaid after kharif breaks the calendar that begins just after spring, and putting kharif first ignores the spring starting point.

10. (D) Sowing with the June monsoon and harvesting as the rains end is the kharif pattern, and kharif crops can live on monsoon rain. Rabi crops grow in the dry winter and usually need irrigation, the reverse timing. Zaid fills the short hot gap, not the monsoon. The last option pairs the wrong season label with the right water source.

11. (A) Kharif crops grow during the rainy monsoon and can live on rain, whereas rabi crops grow in the dry winter when little rain falls, so they need artificial watering. Rabi crops need water like any crop; irrigation supplies it rather than tricking them. The

monsoon falls in the rainy season, not winter. And kharif crops there rely largely on monsoon rain, not canals.

12. (C) The northeast monsoon brings rain mainly from October to December, so that is when she can count on it. June to September is the southwest monsoon, a different system. March to May is the hot pre-monsoon stretch. Indian rainfall is concentrated in monsoon periods, not spread evenly through the year.

13. (B) A rain-fed farm depends on the monsoon as its only water source, so a weak monsoon hurts it most; an irrigated farm can draw on wells and canals and is buffered. Tube wells and canals do not stop working merely because the monsoon is weak. Growing the same crop does not equalise their water security. And less rain cannot help a rain-fed crop grow better.

14. (B) Wheat is matched to the cool, comparatively dry winter and is harvested in spring, so sowing it in the hot, wet monsoon mismatches its needs and the crop suffers. Crops are not season-neutral, which is why kharif, rabi and zaid exist. Renaming it kharif or zaid does not change wheat's growing requirements, and no crop ripens in two weeks.

15. (A) Irrigation supplies water artificially, so it cushions farms against a weak or late monsoon and directly reduces monsoon-driven failures. Banning kharif crops would waste the monsoon's rain. Pushing more reliance on rainfall would increase vulnerability, not reduce it. And monocropping the same crop yearly drains soil nutrients and does nothing for water supply.

16. (C) The short, hot stretch between the rabi harvest and kharif sowing is the zaid season, so a quick crop there is a zaid crop. A rabi crop occupies the cool winter, and a kharif crop is sown with the southwest monsoon, neither of which is the in-between window. A multi-year perennial is the opposite of a quick short-season crop.

17. (D) The false statement misnames rabi: rabi is the cool-winter crop, not the monsoon crop sown in June. The other three are accurate: kharif is sown with the southwest monsoon in the hot rainy season; rabi grows in the cool drier winter and is harvested in spring; and zaid is the short hot in-between crop.

18. (D) Alluvial soil is fertile silt deposited by rivers, carrying nutrients and retaining moisture, which is why it yields more. It is river-borne silt, not wind-blown desert sand. Even fertile soil supplies nutrients to plants, which is precisely why it is fertile. And alluvium makes a productive bed for roots rather than blocking them.

19. (A) Humus is decomposed plant and animal matter; mixing it in supplies nutrients and improves the soil's ability to hold moisture, so the crop does better. It helps retain water rather than removing it, it loosens and enriches soil rather than hardening it into rock, and it supports rather than destroys the living matter that keeps soil healthy.

20. (A) Crop rotation works because different crops use and replenish different nutrients, so no single nutrient is exhausted and the soil stays balanced, which is why the next cereal does better. Rotation means changing the crop grown, not spinning the field. It conserves nutrients rather than stripping them all at once, and it keeps the field productive rather than ending cultivation.

21. (C) Terrace farming cuts a slope into level steps so rainwater is held on each step and soil is not washed away, allowing crops on steep ground. The steps still need soil and water; they manage them better. They conserve soil rather than turning land to sand, and every crop still needs seeds, so terracing cannot remove that need.

22. (A) A purely rain-fed farm depends entirely on the monsoon, so a weak or late monsoon is its chief threat. The other risks all assume irrigation equipment she does not have: there is no canal to flood the field, no pump to break and no drip line to clog. That is exactly why irrigation, where available, makes a farm less vulnerable.

23. (A) A bund is a low earthen embankment that traps water in the field and reduces soil erosion, which is why farmers build them for water and soil management. It is not a seed, because it does not grow; not a chemical, because it does not kill pests; and not a crop, because it is an earthen structure, not a plant.

24. (B) The mix of ancient tanks and bunds with modern canals and wells shows that storing and supplying water for crops has mattered across the ages, addressed by both old and new means. It was not invented in the Green-Revolution era, the very existence of these structures proves farmers thought hard about water, and water management matters in seasonally rained regions too, not only where it never rains.

25. (A) The honest comparison is a trade-off: saved seeds are cheap and locally suited, while bought high-yield seeds can raise yields but bring extra cost and usually more water and chemicals, which is why many farmers blend the two. Saved seeds do not always out-yield modern ones, bought seeds are not free and often demand more inputs, and seeds clearly differ, so the choice does matter.

26. (B) Blending traditional and modern methods lets a farmer earn decent yields while guarding the soil and limiting costs, which is the balance the chapter highlights. The aim is to protect the land, not harm it; there is no law banning all modern tools; and the point is to combine the strengths of both approaches, not to disprove that chemicals can raise yields, which they can.

27. (B) The Green Revolution combined high-yielding seeds, irrigation, chemical fertilisers and pesticides, and machinery; it concerned farming inputs and yields, not how goods were traded. A switch to barter is not part of it, so it is the odd one out. The other three were all core components of the package.

28. (B) The Green Revolution's national-scale success was raising food-grain output so India could feed itself, though it also degraded soil, depleted groundwater, polluted and harmed health. Saved seeds do not always out-produce high-yield seeds. The revolution raised, not lowered, output. And both approaches clearly affect how much food is produced, so 'no effect' is false.

29. (C) Drawing water from underground faster than it is replenished lowers the water table, so heavy pumping depletes groundwater and threatens future supply. Pumping removes water, so it cannot raise the table or flood the region, it clearly does affect groundwater rather than leaving it unchanged, and it cannot manufacture new underground rivers.

30. (C) Output did rise, so the assertion is true, but the claim of no harmful side effects is false: the revolution degraded soil, depleted groundwater, polluted soil and water, harmed health and pushed many farmers into debt. So the reason is wrong while the assertion stands, which rules out the options calling the assertion false or the reason true.

31. (B) The pattern ties together: intensive Green-Revolution methods lifted yields but also drained soil and groundwater and raised input costs, so the same approach produced both the gains and the later troubles. The problems are clearly linked to the method, the region did adopt the revolution, and the debt traces to the high cost of seeds and chemicals, not to low yields.

32. (A) Sustained heavy chemical use pollutes soil and water and wears down the soil's natural fertility, the chain the chapter describes. It does not make soil richer and cleaner, the reverse of what happens; chemicals do not simply vanish without effect; and they have no power to increase rainfall.

33. (B) The accurate verdict is mixed: yields rose, but at the cost of degraded soil, falling groundwater, chemical pollution, health harms and farmer debt. It did not reduce the food supply, since output rose; it clearly had measurable effects; and it did not benefit soil and groundwater, which were among the very things it harmed.

34. (D) The defining difference is that the organic farm shuns synthetic fertilisers and pesticides and uses natural inputs instead. Organic wheat still grows in soil, and the organic farm does produce wheat, so the no-soil and no-wheat claims are wrong. Using more chemical pesticide would make it the chemical-intensive farm, not the organic one.

35. (D) Reducing chemical inputs and heavy irrigation lessens water pollution and slows groundwater depletion, so over time groundwater can improve. Organic farming still uses water and soil, so 'no water, no soil' is wrong; natural inputs do not refill aquifers overnight; and organic crops do grow and use water, just more sustainably.

36. (B) Organic and natural farming are framed as a response because they consciously cut chemical inputs after the Green Revolution showed the damage those inputs caused.

Older traditional farming did not rely on heavy chemicals, the chemical era is the new thing being responded to, organic methods are not a 1960s invention, and natural inputs like manure and neem have been used for ages.

37. (A) Sustainable farming means meeting present needs while protecting soil, water and the capacity to farm in the future. Using up resources as fast as possible, or ignoring the future to chase this year's profit, is the opposite of sustainability, and farming that produces no food today fails the 'meets today's needs' part of the definition.

38. (B) Practices that protect and maintain soil health, such as rotation, adding humus and stopping erosion, are together called soil conservation. Irrigation is about supplying water, not protecting soil; mechanisation refers to using machines, not conserving soil; and exporting soil is not a farming practice at all.

39. (A) A sensible middle path combines natural methods with measured modern inputs, balancing yield, soil health and cost while she transitions, the blended approach the chapter endorses. Piling on chemicals worsens the long-term harm she is escaping, stopping farming sacrifices her livelihood, and ignoring soil health to chase yield repeats the Green Revolution's mistake.

40. (C) The ending '-cide' means 'killer', so a pesticide is a substance that kills crop pests, which corrects the teacher's claim. '-cide' does not mean water, seed or garden, so the other readings of the word are all false.

41. (D) Hydroponics grows plants in nutrient-rich water, feeding minerals straight to the roots, which proves plants can grow without soil. Plants do need nutrients, so 'no nutrients' is false; they are not restricted to bare rock; and they certainly are not limited to deserts.

42. (C) Because hydroponics grows plants in nutrient-rich water rather than soil, it is valuable for urban farming and places where the soil is poor. It needs water, so it cannot work without any water; it supplies nutrients through the water rather than removing the need for them; and it is a modern technique, not one limited to ancient times.

43. (C) The chapter stresses that in rural India most agricultural operations are carried out by women, so a survey of hands-on field work would mostly find women. It is not men only, nor machines only, since much of the labour is manual, and it is local rural labour rather than imported foreign workers.

44. (C) If much of women's work is unpaid and unrecorded, the records understate it, yet women still perform a large share of the real work, so their contribution is undercounted, not small. Unpaid work is still genuine farm labour, so 'only paid male labour counts' is wrong, and farming plainly still depends heavily on human labour.

45. (D) The programme fails because a large share of farm work is done by women and by landless labourers who may own no land, so assuming every farmer is a male owner leaves them out. It is not because women contribute nothing, which is false, nor because all farmers are wealthy owners, and farm labour is still essential, so support for workers matters.

46. (D) The correction is twofold: Indian farming is far from fully mechanised, and women continue to do most of the actual field work, so their role has not ended. Saying machines do everything overstates mechanisation, claiming women never worked the fields contradicts the chapter, and asserting men always did all the work is the very misconception being challenged.

47. (C) If women do most of the work, then aiming training, tools and support at them reaches the people actually farming, which is the sensible response. Withdrawing support from farms with women, or assuming women cannot benefit from better tools, ignores the very workers carrying the load, and replacing them with imported labour neither follows from the fact nor reflects how rural farming operates.

48. (D) Machines speed up heavy tasks and reduce manual labour, most usefully on larger holdings, which is the core benefit of mechanisation. It does not remove the need for soil, seeds or water, which crops still require; it does not guarantee wealth, since machines also cost money; and it does not restrict farming to one season.

49. (C) Heavy machinery fits larger holdings and can be costly or impractical on very small, fragmented plots, so the small farmer's caution is reasonable. Mechanisation is not equally easy and cheap for every farm, it is certainly used in India so it is not impossible, and it reduces rather than wholly removes the need for human labour, which farms still rely on.

50. (A) Earning little despite a good harvest because prices crash is the problem of uncertain market prices, a real challenge for farmers. It is not a weak monsoon, since the harvest was good, nor barren soil, since the field produced well, and the issue is the low price, not a total absence of buyers.

51. (C) Erratic rains, heatwaves and shifting seasons are signs of climate change, which makes the timing of farming unreliable. It is not an excess of irrigation water, nor soil becoming too fertile, which would not disrupt planning, and it is not a sudden loss of farm labour, which is a separate matter.

52. (D) Repeated division shrinks and scatters holdings, which makes efficient farming and the use of machinery harder, a real challenge the chapter notes. It does the opposite of enlarging farms, it does not improve soil fertility, and it cannot guarantee higher crop prices, which depend on the market.

53. (B) Today's farmers face shrinking, fragmented land, climate change, unstable prices, water stress and debt, and naming these is accurate. The other options are unreal: farmers do not enjoy abundant land and perfect prices, they clearly do face difficulties, and India grows a wide range of crops, so 'no crops' is false.

54. (C) The fair lesson is a trade-off across time: chemical-boosted yields give short-term output gains while imposing long-term costs on soil, water and health, exactly what the region shows. Saying it brings only benefits ignores the documented harms, only costs ignores the real rise in output, and it is plainly connected to the soil and water it degraded.

55. (D) Choosing a natural input like neem over a synthetic pesticide is a hallmark of organic farming. The Green-Revolution package is defined by chemical inputs, the reverse of this choice. Selling crops abroad is unrelated to which pesticide is used, and she is still controlling pests, just by natural means rather than abandoning the effort.

56. (B) The accurate summary holds three threads together: the Green Revolution's record was mixed, women do most rural farm work, and many farmers combine old and new methods. Calling the Green Revolution purely good ignores its harms and misnames the typical worker, claiming all farming is identical denies its real variety, and saying women do no work while machines do everything contradicts the chapter on both counts.

57. (B) Farming evidence reaching back to the 7th-8th millennium BCE and persisting to the present shows agriculture is an ancient, continuous Indian tradition. It is therefore not a recent invention, not wholly borrowed from abroad, and certainly not abandoned, since it continues today.

58. (A) India has several major soils, alluvial, black, red and others, suited to different crops, so what grows well varies by place. Saying every field has identical soil restates the teacher's error. Soil type clearly affects which crops thrive, and the variety of soils is precisely why many different crops grow across the country, not just one.

59. (A) Rice grown in the wet monsoon season is a kharif crop, while wheat grown in the cool dry winter is a rabi crop, so the pairing is kharif then rabi. Reversing them mislabels rice as rabi and wheat as kharif. Zaid is the short hot in-between crop and fits neither rice in the monsoon nor wheat in winter.

60. (D) Women are central to rural farming because they do the majority of the physical field tasks, from sowing to harvest. It is not because they own most farmland, since land is often not in their name, nor because they alone run heavy machinery. And they are very much in the fields doing the work, not managing only from a distance.